

4. Tools and Technologies Used

This section explains all the tools and technologies used in the **ReqRes API Testing Project using Postman**. Each tool plays an important role in designing, executing, and automating API tests. These tools together help simulate a real-world API testing environment used in software QA industry.

4.1 Postman

Overview

Postman is the primary tool used in this project for **manual API testing**. It is a widely used API development and testing tool that allows testers to send HTTP requests to servers and analyze responses in a user-friendly interface.

Role in This Project

In this project, Postman is used to:

- Send API requests (GET, POST, PUT, PATCH, DELETE)
- Validate API responses
- Test authentication APIs
- Organize APIs into collections and folders
- Create environments for managing variables (like base URL)
- Write test scripts using JavaScript

Key Features Used

- **Collections** → Grouping all ReqRes APIs in one structure
- **Environments** → Storing variables like `base_url`
- **Test Scripts** → Writing validations for status codes and response data
- **Collection Runner** → Running multiple APIs together in sequence

Importance

Postman helps testers manually verify APIs without writing full automation code. It is widely used in QA teams for quick debugging and validation of APIs.

4.2 Newman

Overview

Newman is the **command-line tool for Postman**. It allows execution of Postman collections directly from the terminal or command prompt.

Role in This Project

In this project, Newman is used to:

- Run Postman collections outside the Postman UI
- Execute all API test cases in bulk
- Automate API testing process
- Generate test execution reports

Key Features Used

- Command-line execution of collections
- Integration with Postman JSON collection file
- HTML and CLI report generation
- Useful for CI/CD pipeline integration

Example Use

Instead of manually clicking “Run” in Postman, Newman allows:

```
newman run collection.json
```

Importance

Newman is important for **automation testing**, as it helps simulate real-world DevOps and CI/CD workflows where tests run automatically after code deployment.

4.3 Node.js

Overview

Node.js is a JavaScript runtime environment used to execute JavaScript outside the browser.

Role in This Project

Node.js is used to:

- Install and run Newman using npm (Node Package Manager)
- Provide runtime environment for command-line tools
- Support execution of API automation scripts

Installation Use

Newman is installed using Node.js:

```
npm install -g newman
```

Key Features Used

- Package management using npm
- Runtime support for Newman
- Lightweight and fast execution environment

Importance

Node.js acts as a base platform that enables Newman to run. Without Node.js, Newman cannot be installed or executed.

4.4 ReqRes REST API

Overview

ReqRes (<https://reqres.in>) is a **public REST API service** used for testing and learning API automation.

It provides fake but realistic API endpoints for user management operations like login, user retrieval, creation, update, and deletion.

Role in This Project

ReqRes API is the **system under test (SUT)** in this project. All testing activities are performed on its endpoints.

APIs Used

- **Login API** → Authentication testing
- **Get Users API** → Fetch list of users
- **Get Single User API** → Fetch specific user
- **Create User API** → Add new user
- **Update User API (PUT/PATCH)** → Modify user data
- **Delete User API** → Remove user

Key Features

- Free and publicly available
- No setup required
- Returns JSON responses
- Supports all HTTP methods
- Ideal for beginners and QA practice

Importance

ReqRes provides a safe environment to learn API testing without affecting real production systems. It simulates real-world backend behavior.

4.5 Summary of Tools

Tool	Purpose
Postman	Manual API testing, request execution, validation
Newman	Automated collection execution via CLI
Node.js	Runtime environment for Newman
ReqRes API Test server	providing REST API endpoints

4.6 Final Conclusion of Tools Section

The combination of Postman, Newman, Node.js, and ReqRes API provides a complete environment for learning and executing API testing. Postman is used for manual testing, Newman enables automation, Node.js supports execution, and ReqRes serves as the test backend system.

Together, these tools simulate a real-world API testing framework used in software development and QA industries.
